

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

ANS:-

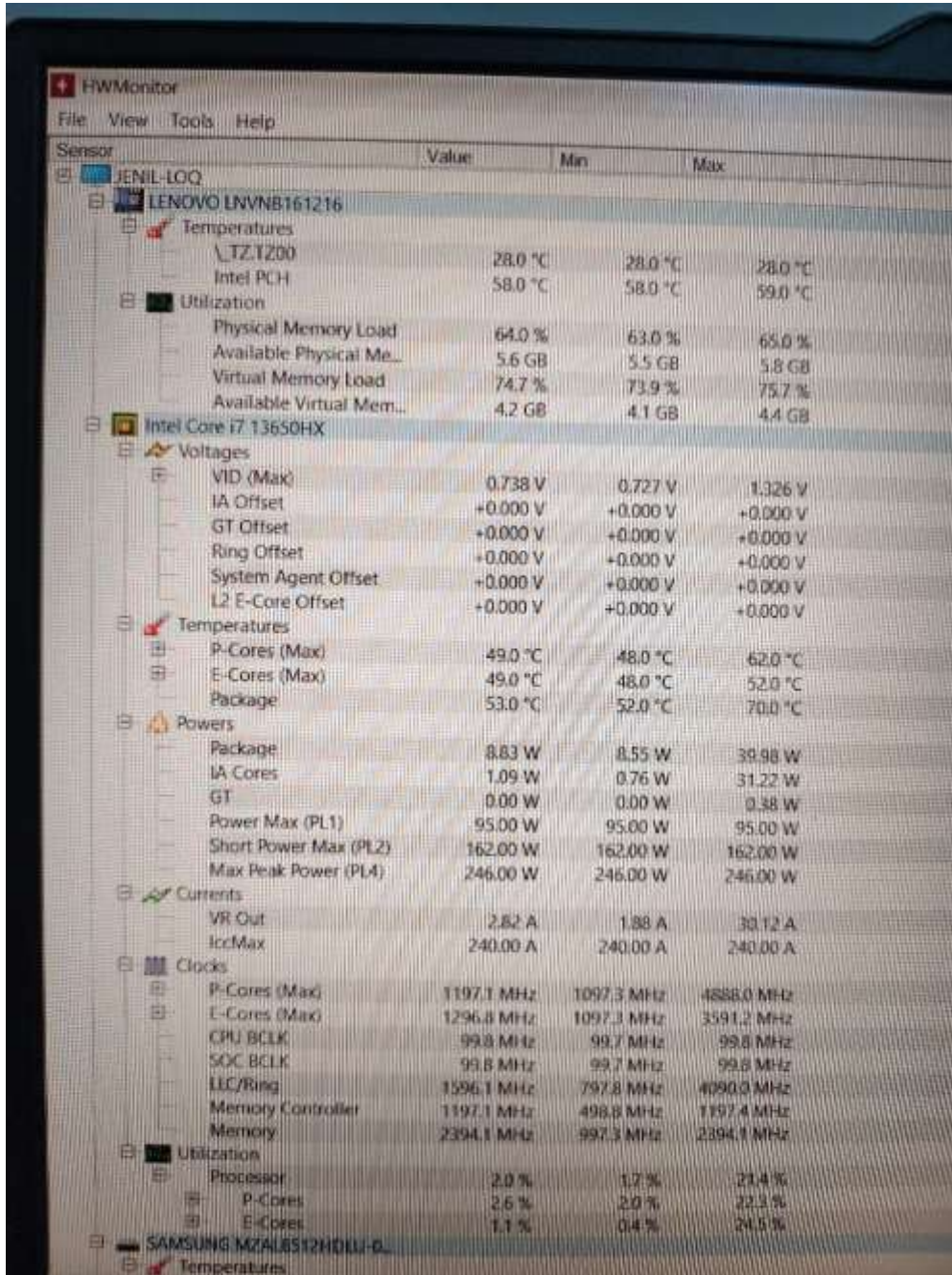
1. **Testing a Laptop DC Power Jack / Charger:** The multimeter measures the DC output voltage at the charger pin to verify if the adapter is delivering the required voltage to the laptop.
2. **Checking Smartphone Board-Level Fuses and Rails:** The multimeter tests continuity across micro-fuses and capacitors to detect short circuits or broken traces that prevent the phone from turning on.
3. **Verifying Desktop PSU Output Voltages:** The multimeter measures the standard 3.3V, 5V, and 12V rails on ATX connectors to confirm whether the power supply is delivering stable electrical power under load.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Hint: Focus on the codes shown and how they relate to hardware issues.

ANS:- A POST (Power-On Self-Test) expansion card plugs into a PCI/PCIe or LPC slot and displays two-digit or four-digit hexadecimal diagnostic codes in real time during system boot. As the BIOS initializes each hardware component, the on-board LED display cycles through sequential hex codes. If the system fails to power on or display an image, the POST card stops at a specific error code (such as freezing on memory initialization or CPU test routines), allowing technicians to immediately isolate a faulty RAM stick, corrupt BIOS, or bad motherboard trace without trial-and-error guessing.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.

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The screenshot shows the HWMonitor application window. The title bar reads 'HWMonitor'. The menu bar includes 'File', 'View', 'Tools', and 'Help'. The main content area displays a tree view of sensors. The 'Sensors' folder is expanded, showing 'JENIL-LOQ' and 'LENOVO LNVNB161216'. Under 'LENOVO LNVNB161216', the following categories are visible: 'Temperatures' (with sub-items for \TZ.TZ00, Intel PCH, and a list of core temperatures), 'Utilization' (Physical Memory Load, Available Physical Me..., Virtual Memory Load, Available Virtual Mem...), 'Intel Core i7 13650HX' (with sub-items for Voltages, Temperatures, Powers, Currents, Clocks, and Utilization), and 'SAMSUNG MZAL6S12HDLU-D' (with a sub-item for Temperatures). The 'Temperatures' section for the Intel Core i7 13650HX is expanded, showing a table of values, minimums, and maximums for various sensors.

Sensor	Value	Min	Max
LENOVO LNVNB161216			
Temperatures			
\TZ.TZ00	28.0 °C	28.0 °C	28.0 °C
Intel PCH	58.0 °C	58.0 °C	59.0 °C
Utilization			
Physical Memory Load	64.0 %	63.0 %	65.0 %
Available Physical Me...	5.6 GB	5.5 GB	5.8 GB
Virtual Memory Load	74.7 %	73.9 %	75.7 %
Available Virtual Mem...	4.2 GB	4.1 GB	4.4 GB
Intel Core i7 13650HX			
Voltages			
VID (Max)	0.738 V	0.727 V	1.326 V
IA Offset	+0.000 V	+0.000 V	+0.000 V
GT Offset	+0.000 V	+0.000 V	+0.000 V
Ring Offset	+0.000 V	+0.000 V	+0.000 V
System Agent Offset	+0.000 V	+0.000 V	+0.000 V
L2 E-Core Offset	+0.000 V	+0.000 V	+0.000 V
Temperatures			
P-Cores (Max)	49.0 °C	48.0 °C	62.0 °C
E-Cores (Max)	49.0 °C	48.0 °C	52.0 °C
Package	53.0 °C	52.0 °C	70.0 °C
Powers			
Package	8.83 W	8.55 W	39.98 W
IA Cores	1.09 W	0.76 W	31.22 W
GT	0.00 W	0.00 W	0.38 W
Power Max (PL1)	95.00 W	95.00 W	95.00 W
Short Power Max (PL2)	162.00 W	162.00 W	162.00 W
Max Peak Power (PL4)	246.00 W	246.00 W	246.00 W
Currents			
VR Out	2.82 A	1.88 A	30.12 A
IccMax	240.00 A	240.00 A	240.00 A
Clocks			
P-Cores (Max)	1197.1 MHz	1097.3 MHz	4888.0 MHz
E-Cores (Max)	1296.8 MHz	1097.3 MHz	3591.2 MHz
CPU BCLK	99.8 MHz	99.7 MHz	99.8 MHz
SOC BCLK	99.8 MHz	99.7 MHz	99.8 MHz
LLC/Ring	1596.1 MHz	797.8 MHz	4090.0 MHz
Memory Controller	1197.1 MHz	498.8 MHz	1197.4 MHz
Memory	2394.1 MHz	997.3 MHz	2394.1 MHz
Utilization			
Processor	2.0 %	1.7 %	21.4 %
P-Cores	2.6 %	2.0 %	23.3 %
E-Cores	1.1 %	0.4 %	24.5 %
SAMSUNG MZAL6S12HDLU-D			
Temperatures			

I downloaded and ran **HWMonitor** on my laptop to monitor live hardware diagnostics and system metrics. The main dashboard displays critical telemetry, including individual CPU core temperatures, fan speeds, core voltage levels, memory utilization, and storage drive health parameters. Checking

these real-time operational values helps detect thermal throttling, failing cooling fans, or voltage instabilities before they lead to permanent hardware damage.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

ANS:-

Tool	What it checks	Example Problem Detected
Multimeter	Measures electrical values such as voltage, resistance, and continuity.	Faulty power supply or a broken cable connection.
POST Card	Checks the computer's Power-On Self-Test (POST) process and displays diagnostic codes during startup.	Faulty RAM or CPU/motherboard initialization failure.
Software Diagnostics	Uses diagnostic programs to check hardware, drivers, operating system, and system performance.	Hard-drive errors , outdated drivers, or excessive CPU/RAM usage.